

EE3621: Digital Signal Processing

Student Lecture Notes — Lecture 21

Linear Phase FIR Filters, Characteristics, Zeros & Moving Average Filters

III B.Tech EEE — Core Exam Preparation Guide

1 Core Definitions & Linear Phase Conditions

1.1 Distortionless Transmission & Phase Delay vs. Group Delay

An LTI filter transmits an input signal without waveform distortion if all frequency components experience:

1. A constant gain: $|H(e^{j\omega})| = A_0$
 2. A strictly constant time delay: $\tau_g = \alpha$
- **Phase Delay** (carrier time delay) and **Group Delay** (envelope time delay):

$$\tau_p(\omega) = -\frac{\theta(\omega)}{\omega}, \quad \tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega} \quad (1)$$

The Linear Phase Criterion: If $\theta(\omega) = -\alpha\omega$, then $\tau_p(\omega) = \tau_g(\omega) = \alpha = \text{constant}$. This eliminates phase distortion and envelope dispersion.

1.2 Time-Domain Conditions for Linear Phase

For a causal FIR filter of length N (coefficients $h[0], h[1], \dots, h[N-1]$):

- **Strict Linear Phase** ($\theta(\omega) = -\alpha\omega$): Requires **even symmetry**:

$$h[n] = h[N-1-n], \quad \alpha = \frac{N-1}{2} \quad (2)$$

- **Generalized Linear Phase** ($\theta(\omega) = \beta - \alpha\omega$, where $\beta = \pm\pi/2$): Requires **odd antisymmetry**:

$$h[n] = -h[N-1-n], \quad \alpha = \frac{N-1}{2} \quad (3)$$

2 The 4 Linear-Phase FIR Filter Types (Master Reference)

Every linear-phase FIR filter belongs to one of four distinct classes based on coefficient symmetry and length parity:

Table 1: Classification and Filter Suitability of Linear-Phase FIR Filters

Type	Symmetry	Length N	Phase $\theta(\omega)$	Mandatory Zeros	Filter Suitability
Type 1	Symmetric ($h[n] = h[N-1-n]$)	Odd (M even)	$-\alpha\omega$	None	Universal: LPF, HPF, BPF, BSF
Type 2	Symmetric ($h[n] = h[N-1-n]$)	Even (M odd)	$-\alpha\omega$	$\omega = \pi$ ($z = -1$)	LPF, BPF only (Never HPF/BSF!)
Type 3	Antisymmetric ($h[n] = -h[N-1-n]$)	Odd (M even)	$\frac{\pi}{2} - \alpha\omega$	$\omega = 0, \pi$ ($z = \pm 1$)	BPF only (Never LPF/HPF/BSF)
Type 4	Antisymmetric ($h[n] = -h[N-1-n]$)	Even (M odd)	$\frac{\pi}{2} - \alpha\omega$	$\omega = 0$ ($z = 1$)	HPF, BPF, Differentiator

Critical Exam Insights:

- **Why Type 2 cannot be Highpass:** At $\omega = \pi$, $e^{j\pi(n-\alpha)} = \cos(\pi(n-\alpha))$. For N even, $\alpha = (N-1)/2$ is a half-integer, making $\cos(\pi(n-\alpha)) = 0$. Hence, $\mathbf{H}(\mathbf{e}^{j\pi}) \equiv \mathbf{0}$ unconditionally!
- **Why Antisymmetric Filters (Types 3 & 4) have zero at $\omega = 0$:** Since $h[n] = -h[N-1-n]$, setting $\omega = 0$ gives $H(e^{j0}) = \sum h[n] = 0$. They block DC completely.

3 Location of Zeros in the Complex Z-Plane

Linear phase imposes reciprocal symmetry on the system function:

$$H(z) = \pm z^{-(N-1)} H(z^{-1}) \quad (4)$$

Consequently, if $z_0 = re^{j\phi}$ is a zero of $H(z)$, its reciprocal $1/z_0 = \frac{1}{r}e^{-j\phi}$ must also be a zero. For real impulse responses, complex zeros must also occur in conjugate pairs.

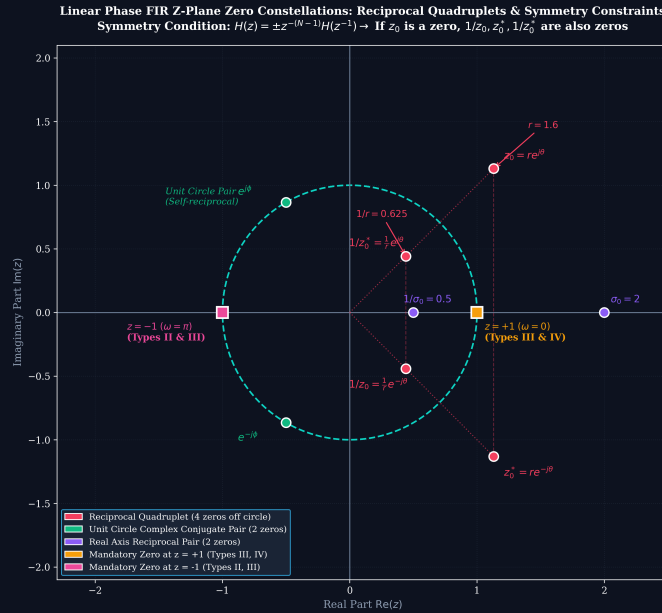


Figure 1: Zero constellations for linear-phase FIR systems: complex quadruplets off the unit circle, pairs on the unit circle ($|z| = 1$), and mandatory boundary zeros at $z = \pm 1$.

- **Complex zero off the unit circle ($|z| \neq 1$):** Appears as a **reciprocal conjugate quadruplet of 4 zeros**:

$$\left\{ re^{j\phi}, \quad re^{-j\phi}, \quad \frac{1}{r}e^{j\phi}, \quad \frac{1}{r}e^{-j\phi} \right\} \quad (5)$$

- **Complex zero on the unit circle ($|z| = 1$):** Since $1/e^{j\phi} = e^{-j\phi}$, the zero is its own reciprocal. It occurs as a **conjugate pair of 2 zeros**:

$$\left\{ e^{j\phi}, \quad e^{-j\phi} \right\} \quad (6)$$

These zeros create sharp transmission nulls in the filter's stopband.

- **Real zero off the unit circle:** Occurs as a **reciprocal pair of 2 zeros**: $\{\sigma, 1/\sigma\}$.
- **Real zero on the unit circle:** A single zero at $z = 1$ ($\omega = 0$) or $z = -1$ ($\omega = \pi$).

4 Moving Average (MA) Filters

4.1 Definition and Impulse Response

An M -point Moving Average filter computes the local arithmetic mean of the input over a sliding window of length M :

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n-k] = \frac{1}{M} (x[n] + x[n-1] + \cdots + x[n-M+1]) \quad (7)$$

Its unit impulse response is a causal rectangular sequence of amplitude $1/M$:

$$h[n] = \begin{cases} \frac{1}{M}, & 0 \leq n \leq M-1 \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

Linear Phase Property: Because $h[n] = h[M-1-n] = \frac{1}{M}$ is perfectly symmetric about $\alpha = \frac{M-1}{2}$:

- If M is odd $\implies$ **Type 1 FIR Filter** with integer group delay $\tau = \frac{M-1}{2}$ samples.
- If M is even $\implies$ **Type 2 FIR Filter** with half-integer group delay $\tau = \frac{M-1}{2}$ samples.

4.2 Step-by-Step Derivation of Dirichlet Sinc Frequency Response

1. **Apply the DTFT Definition:**

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n} = \frac{1}{M} \sum_{n=0}^{M-1} e^{-j\omega n} = \frac{1}{M} \sum_{n=0}^{M-1} (e^{-j\omega})^n \quad (9)$$

2. **Sum the Finite Geometric Series:** Recall the geometric series sum $\sum_{n=0}^{M-1} r^n = \frac{1-r^M}{1-r}$ with common ratio $r = e^{-j\omega} \neq 1$:

$$H(e^{j\omega}) = \frac{1}{M} \left[\frac{1 - (e^{-j\omega})^M}{1 - e^{-j\omega}} \right] = \frac{1}{M} \left[\frac{1 - e^{-jM\omega}}{1 - e^{-j\omega}} \right] \quad (10)$$

3. **Factor Out Half-Angle Phase Terms (Symmetric Extraction):** Factor out $e^{-jM\omega/2}$ from the numerator and $e^{-j\omega/2}$ from the denominator:

$$1 - e^{-jM\omega} = e^{-j\frac{M\omega}{2}} \left(e^{j\frac{M\omega}{2}} - e^{-j\frac{M\omega}{2}} \right), \quad 1 - e^{-j\omega} = e^{-j\frac{\omega}{2}} \left(e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}} \right) \quad (11)$$

4. **Apply Euler's Sine Formula ($e^{j\theta} - e^{-j\theta} = 2j \sin \theta$) & Simplify:** Cancelling $2j$ and combining the linear phase factor $\frac{e^{-jM\omega/2}}{e^{-j\omega/2}} = e^{-j(\frac{M-1}{2})\omega}$:

$$H(e^{j\omega}) = \frac{1}{M} \frac{e^{-j\frac{M\omega}{2}} \cdot 2j \sin \left(\frac{M\omega}{2} \right)}{e^{-j\frac{\omega}{2}} \cdot 2j \sin \left(\frac{\omega}{2} \right)} = \left[\frac{\sin \left(\frac{M\omega}{2} \right)}{M \sin \left(\frac{\omega}{2} \right)} \right] e^{-j(\frac{M-1}{2})\omega} \quad (12)$$

4.3 Key Mathematical Characteristics & Transmission Nulls

- **Real Amplitude Response:** $A(\omega) = \frac{\sin(M\omega/2)}{M \sin(\omega/2)}$, with phase $\theta(\omega) = -(\frac{M-1}{2})\omega$.
- **Continuous vs. Discrete Sinc:** For small $\omega \ll 1$, $\sin(\omega/2) \approx \omega/2$, so $A(\omega) \approx \frac{\sin(M\omega/2)}{M(\omega/2)} = \text{sinc} \left(\frac{M\omega}{2\pi} \right)$. The discrete sine denominator makes $A(\omega)$ strictly 2π -periodic.
- **DC Gain ($\omega = 0$):** By L'Hôpital's rule, $A(0) = \lim_{\omega \rightarrow 0} \frac{\frac{M}{2} \cos(M\omega/2)}{M \cdot \frac{1}{2} \cos(\omega/2)} = 1$ (unity DC gain).

- **Stopband Transmission Nulls (Unit Circle Zeros):** $A(\omega) = 0$ when $\sin(M\omega/2) = 0$ and $\sin(\omega/2) \neq 0$:

$$\frac{M\omega_k}{2} = k\pi \implies \omega_k = \frac{2\pi k}{M}, \quad k = 1, 2, \dots, M-1 \quad (13)$$

In the Z -domain, $H(z) = \frac{1}{M} \frac{1-z^{-M}}{1-z^{-1}}$. The root at $z = 1$ ($k = 0$) is cancelled by the pole at $z = 1$, leaving exactly $M-1$ zeros on the unit circle: $\mathbf{z}_k = e^{j2\pi k/M}$ ($k \neq 0$).

5 Standard Solved Exam Example

Problem: An FIR filter has the impulse response:

$$h[n] = \{1, 2, 3, 2, 1\}, \quad \text{for } n = 0, 1, 2, 3, 4 \quad (14)$$

1. Verify linear phase and determine the group delay τ_g .
2. Identify the filter type (Type 1 to 4) and its filter suitability.
3. **Derive the DTFT step-by-step from first principles (no shortcuts)**, obtain the amplitude response $A(\omega)$ and phase response $\theta(\omega)$, and evaluate $A(\omega)$ at $\omega = 0, \frac{\pi}{2}, \frac{2\pi}{3}, \pi$.
4. Find all zeros of $H(z)$ algebraically and verify their constellation on the unit circle.

Solution:

Step 1: Symmetry and Group Delay

- Length $N = 5$ (Odd length). Order $M = N - 1 = 4$ (Even order).
- Center of symmetry: $\alpha = \frac{N-1}{2} = \frac{5-1}{2} = 2$.
- Symmetry verification:

$$h[0] = h[4] = 1, \quad h[1] = h[3] = 2, \quad h[2] = 3 \quad (15)$$

Since $h[n] = h[N-1-n] = h[4-n]$ for all $n \in \{0, 1, 2, 3, 4\}$, the filter possesses **even symmetry**.

- The group delay is strictly constant:

$$\tau_g = \alpha = 2 \text{ samples} \quad (\text{independent of frequency } \omega) \quad (16)$$

Step 2: Filter Classification

- **Classification:** Even symmetry + Odd length $N = 5 \implies$ **Type 1 Linear-Phase FIR Filter**.
- **Filter Suitability:** Universal. Type 1 filters have no mandatory zeros at either $\omega = 0$ ($z = 1$) or $\omega = \pi$ ($z = -1$). Hence, they can freely synthesize Lowpass, Highpass, Bandpass, or Bandstop filters.

Step 3: Step-by-Step Derivation of DTFT & Amplitude Response (First Principles)

1. **Fundamental Definition of the DTFT:**

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n} = \sum_{n=0}^4 h[n]e^{-j\omega n} \quad (17)$$

2. **Expand Sample-by-Sample for all $n \in \{0, 1, 2, 3, 4\}$:**

$$H(e^{j\omega}) = h[0]e^{-j\omega(0)} + h[1]e^{-j\omega(1)} + h[2]e^{-j\omega(2)} + h[3]e^{-j\omega(3)} + h[4]e^{-j\omega(4)} \quad (18)$$

3. **Substitute the Numerical Coefficients of $h[n]$:**

$$\begin{aligned} H(e^{j\omega}) &= 1 \cdot e^0 + 2 \cdot e^{-j\omega} + 3 \cdot e^{-j2\omega} + 2 \cdot e^{-j3\omega} + 1 \cdot e^{-j4\omega} \\ &= 1 + 2e^{-j\omega} + 3e^{-j2\omega} + 2e^{-j3\omega} + e^{-j4\omega} \end{aligned} \quad (19)$$

4. **Factor Out the Center Symmetry Delay $e^{-j\alpha\omega} = e^{-j2\omega}$:**

$$H(e^{j\omega}) = e^{-j2\omega} \left[\frac{1}{e^{-j2\omega}} + \frac{2e^{-j\omega}}{e^{-j2\omega}} + \frac{3e^{-j2\omega}}{e^{-j2\omega}} + \frac{2e^{-j3\omega}}{e^{-j2\omega}} + \frac{e^{-j4\omega}}{e^{-j2\omega}} \right] \quad (20)$$

$$H(e^{j\omega}) = e^{-j2\omega} [1 \cdot e^{j2\omega} + 2 \cdot e^{j\omega} + 3 + 2 \cdot e^{-j\omega} + 1 \cdot e^{-j2\omega}] \quad (21)$$

5. **Group Symmetric Conjugate Pairs:**

$$H(e^{j\omega}) = e^{-j2\omega} [3 + 2(e^{j\omega} + e^{-j\omega}) + 1(e^{j2\omega} + e^{-j2\omega})] \quad (22)$$

6. **Apply Euler's Cosine Identity ($e^{j\theta} + e^{-j\theta} = 2 \cos \theta$):**

$$e^{j\omega} + e^{-j\omega} = 2 \cos(\omega), \quad e^{j2\omega} + e^{-j2\omega} = 2 \cos(2\omega) \quad (23)$$

7. **Substitute into the Grouped Expression:**

$$\begin{aligned} H(e^{j\omega}) &= e^{-j2\omega} [3 + 2(2 \cos(\omega)) + 1(2 \cos(2\omega))] \\ &= e^{-j2\omega} [3 + 4 \cos(\omega) + 2 \cos(2\omega)] \end{aligned} \quad (24)$$

8. **Identify Amplitude and Phase Functions:** Comparing with the standard linear-phase polar form $H(e^{j\omega}) = A(\omega)e^{j\theta(\omega)}$:

$$A(\omega) = 3 + 4 \cos(\omega) + 2 \cos(2\omega) \quad (\text{Real Amplitude Response}) \quad (25)$$

$$\theta(\omega) = -2\omega \quad (\text{Strict Linear Phase}) \quad (26)$$

$$\tau_p(\omega) = \tau_g(\omega) = 2 \text{ samples} \quad (\text{Constant Phase \& Group Delay}) \quad (27)$$

9. **Frequency Response Evaluations:**

- **DC Gain ($\omega = 0$):** $A(0) = 3 + 4(1) + 2(1) = 9 = \sum_{n=0}^4 h[n]$.
- **Quarter-Sampling ($\omega = \pi/2$):** $A(\pi/2) = 3 + 4(0) + 2(-1) = 1$.
- **Nyquist Frequency ($\omega = \pi$):** $A(\pi) = 3 + 4(-1) + 2(1) = 3 - 4 + 2 = 1$.
- **Notch Frequency ($\omega = 2\pi/3 = 120^\circ$):** $A(2\pi/3) = 3 + 4(-0.5) + 2(-0.5) = 3 - 2 - 1 = 0$ (Transmission notch).

Step 4: Step-by-Step Algebraic Zero Extraction

1. **Formulate Z-transform & Clear Negative Powers:**

$$H(z) = 1 + 2z^{-1} + 3z^{-2} + 2z^{-3} + z^{-4} = 0 \implies z^4 + 2z^3 + 3z^2 + 2z + 1 = 0 \quad (28)$$

Because coefficients $\{1, 2, 3, 2, 1\}$ are symmetric, this is a **palindromic (reciprocal) polynomial**.

2. **Divide by Center Term z^2 & Group Reciprocal Pairs:**

$$\left(z^2 + \frac{1}{z^2}\right) + 2\left(z + \frac{1}{z}\right) + 3 = 0 \quad (29)$$

3. **Substitute $u = z + z^{-1}$ (where $z^2 + z^{-2} = u^2 - 2$):**

$$(u^2 - 2) + 2u + 3 = 0 \implies u^2 + 2u + 1 = 0 \implies (u + 1)^2 = 0 \implies \mathbf{u = -1} \text{ (multiplicity 2)} \quad (30)$$

4. **Solve for z from $z+z^{-1} = -1 \implies z^2+z+1 = 0$:** Using the quadratic formula $z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$:

$$z = \frac{-1 \pm \sqrt{1-4}}{2} = -\frac{1}{2} \pm j\frac{\sqrt{3}}{2} \quad (31)$$

5. **Convert to Polar Form $re^{j\phi}$:**

$$|z| = \sqrt{(-1/2)^2 + (\sqrt{3}/2)^2} = 1, \quad \phi = \pm 120^\circ = \pm \frac{2\pi}{3} \text{ rad} \quad (32)$$

Therefore, the 4 zeros of $H(z)$ are:

$$\mathbf{z}_{1,2} = \mathbf{e}^{j\frac{2\pi}{3}} \quad (\text{double zero at } +120^\circ), \quad \mathbf{z}_{3,4} = \mathbf{e}^{-j\frac{2\pi}{3}} \quad (\text{double zero at } -120^\circ) \quad (33)$$

6. **Verification of Linear-Phase Zero Property:** All 4 zeros lie directly on the unit circle ($|z| = 1$) at angles $\pm 120^\circ$. For zeros on the unit circle, the reciprocal partner is the complex conjugate: $\frac{1}{e^{j2\pi/3}} = e^{-j2\pi/3}$. Thus, the reciprocal zero pairing theorem is verified. Furthermore, setting $\omega = \pm 2\pi/3$ corresponds exactly to the notch frequency $A(2\pi/3) = 0$ derived in Step 3!

6 Essential Exam Checklist & Pitfalls

- ☐ **Group Delay Formula:** Always use $\tau = \frac{N-1}{2}$, **NOT** $N/2$.
- ☐ **Highpass Filter Rule:** Never select an even-length symmetric filter (Type 2) for a Highpass filter because $H(e^{j\pi}) \equiv 0$.
- ☐ **DC Zero Rule:** Never select an antisymmetric filter (Type 3 or Type 4) for a Lowpass filter because $H(e^{j0}) \equiv 0$.
- ☐ **Reciprocal Zero Pairing:** Always ensure that if z_0 is a zero, $1/z_0$ and z_0^* are also zeros.